

CHICHIBU, Japan

WMO: 476410

Lat: 35.9900N

Lon: 139.0742E

Elev: 234

StdP: 98.55

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.8	-4.7	-14.7	1.1	4.7	-13.0	1.3	4.2	7.0	6.0	6.3	6.3	0.9	220	0.365

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.8	34.4	25.4	32.9	25.0	31.3	24.3	26.3	32.2	25.7	31.2	25.1	29.8	2.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.7	20.3	28.8	24.2	19.7	28.2	23.7	19.1	27.6	83.4	32.4	80.8	31.5	78.1	29.9	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.8	5.0	4.3	-7.7	36.7	0.7	1.0	-8.2	37.4	-8.7	37.9	-9.1	38.5	-9.6	39.2		
(5)			-8.4	27.5	0.7	0.6	-8.9	27.9	-9.4	28.2	-9.8	28.5	-10.3	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.1	2.4	3.4	6.8	12.5	17.6	21.1	25.2	26.3	22.0	16.1	9.8	4.7		
	DBStd	8.71	2.03	2.54	3.16	3.55	2.84	2.69	2.96	2.61	3.15	2.98	2.96	2.55		
	HDD10.0	752	236	186	109	16	0	0	0	0	0	1	40	164		
	HDD18.3	2264	494	419	356	177	46	7	0	0	6	82	256	421		
	CDD10.0	2233	0	1	11	91	237	332	472	504	360	189	34	2		
	CDD18.3	704	0	0	0	2	25	89	214	246	116	13	0	0		
	CDH23.3	5640	0	0	1	44	276	576	1808	2176	702	58	0	0		
	CDH26.7	2151	0	0	0	7	57	158	776	933	212	8	0	0		
(14) Wind	WSAvg	1.5	1.6	1.7	1.8	1.8	1.6	1.5	1.4	1.4	1.3	1.2	1.2	1.4		
(15) Precipitation	PrecAvg	1426	46	41	76	97	123	162	174	183	229	179	72	45		
	PrecMax	2038	122	91	142	287	269	441	345	486	444	548	318	143		
	PrecMin	922	6	4	25	29	48	60	66	30	40	24	6	8		
	PrecStd	228	35	24	31	62	55	74	79	119	126	128	58	30		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	17.5	22.0	27.1	30.7	32.8	36.0	36.1	33.0	27.8	21.2	16.4		
		MCWB	6.9	10.0	12.4	16.3	19.2	23.1	25.7	25.7	24.5	20.7	14.3	9.5		
	2%	DB	11.3	13.6	18.6	24.3	28.2	30.2	34.5	34.6	30.9	24.8	19.2	14.1		
		MCWB	4.8	6.8	10.3	14.3	17.8	21.5	25.5	25.5	23.7	18.5	13.1	8.1		
	5%	DB	9.6	11.3	16.1	22.1	26.4	28.5	32.9	33.3	29.1	22.9	17.4	12.2		
		MCWB	3.6	5.1	8.9	13.6	17.1	20.7	25.1	25.3	23.0	17.4	11.7	6.9		
	10%	DB	8.1	9.4	13.7	20.0	24.5	26.7	31.1	31.8	27.4	21.2	15.6	10.5		
		MCWB	2.7	3.9	7.8	12.6	16.5	20.1	24.5	24.9	22.0	16.6	11.2	5.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.2	10.9	13.5	18.4	21.5	25.0	27.1	27.2	25.7	21.9	16.3	11.6		
		MCDB	11.5	16.4	19.4	23.3	26.9	30.3	33.3	33.4	30.8	26.1	19.1	14.1		
	2%	WB	5.8	7.8	11.6	16.4	19.8	23.5	26.2	26.4	24.7	20.2	14.7	9.3		
		MCDB	9.7	12.3	16.2	22.0	25.2	27.7	32.3	32.4	29.0	23.0	17.0	12.3		
	5%	WB	4.4	6.1	10.0	14.9	18.7	22.4	25.7	25.9	23.9	18.9	13.3	7.8		
		MCDB	8.4	9.9	14.7	19.2	23.4	26.1	31.2	31.6	27.6	21.3	15.7	11.0		
	10%	WB	3.2	4.5	8.5	13.7	17.7	21.4	25.0	25.3	23.1	17.8	12.1	6.5		
		MCDB	7.2	8.4	12.7	18.1	22.3	24.7	29.8	30.3	26.1	20.0	14.4	9.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.0	11.9	12.0	12.3	11.3	8.8	8.6	8.8	8.1	9.0	11.1	12.0		
		MCDBR	14.5	15.3	16.1	16.0	15.5	12.7	12.1	11.8	10.8	11.3	13.3	14.0		
	5% WB	MCWBR	8.6	9.0	8.9	7.6	6.7	5.2	4.5	4.3	4.6	5.5	7.5	8.3		
		MCWBR	13.5	13.8	14.1	12.9	12.3	9.8	10.9	10.5	9.3	9.1	10.6	12.4		
(40) Clear-Sky Solar Irradiance	taub		0.356	0.402	0.467	0.522	0.555	0.599	0.588	0.589	0.513	0.466	0.397	0.357		
	taud		2.253	2.120	1.976	1.892	1.868	1.827	1.910	1.897	2.062	2.113	2.240	2.304		
	Ebn at Noon		820	819	795	773	754	719	724	712	747	743	765	794		
	Edh at Noon		104	134	170	194	202	210	192	191	154	134	106	94		
(44) All-Sky Solar Radiation	RadAvg		2.66	3.30	4.13	4.81	5.19	4.60	4.73	4.74	3.79	3.12	2.56	2.35		
	RadStd		0.15	0.25	0.25	0.44	0.53	0.38	0.73	0.46	0.33	0.32	0.22	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (43 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page